

DISEASES CAUSED BY EXFOLIATIVE TOXINS: STAPHYLOCOCCAL SCALDED- SKIN SYNDROME

Staphylococcus aureus-mediated production of exfoliative toxin results in a spectrum of blistering skin disorders, ranging from localized **bullous impetigo** (see Chap. 177) to the severe generalized form known as **staphylococcal scalded-skin syndrome (SSSS)**. SSSS is a generalized exanthematous disease characterized by cutaneous tenderness, widespread superficial blistering, and epidermal denudation.

Etiology and Pathogenesis

Exfoliative toxins (ETs) are produced by certain strains of *S. aureus*, typically phage group II. The key toxins involved are **exfoliatin A (ETA)** and **exfoliatin B (ETB)**. These are serine proteases that specifically cleave desmoglein-1, a desmosomal cell adhesion protein in the granular layer of the epidermis. This cleavage leads to loss of keratinocyte cohesion, resulting in intraepidermal splitting and blister formation.

GRAM-POSITIVE INFECTIONS ASSOCIATED WITH TOXIN PRODUCTION

AT A GLANCE

Caused by toxin-producing *Staphylococcus aureus* and/or **group A streptococcus** (*Streptococcus pyogenes*).

Three characteristic syndromes, all with skin manifestations:

Predominantly Cutaneous

- **Local:** Bullous impetigo
- **Generalized:** Staphylococcal scalded skin syndrome (SSSS)

Cutaneous and Systemic (Intermediate)

- **Local:** Recurrent toxin-mediated erythema (e.g., recurrent perineal erythema)
- **Generalized:**
 - Recalcitrant erythematous desquamating disorder
 - Other abortive hybrids of predominantly systemic manifestations

Predominantly Systemic

- **Generalized:**
 - **Toxic shock syndrome (TSS)** — caused predominantly by *Staphylococcus aureus* or *Streptococcus pyogenes*
 - **Scarlet fever** — caused predominantly by *Streptococcus pyogenes*, occasionally by *S. aureus*

TABLE 178-1 Toxins and Diseases Due to *Staphylococcus aureus* and Group A *Streptococcus*

BACTERIA	TOXIN	TOXIN TYPE	CLINICAL DISEASE
<i>Staphylococcus aureus</i>	Exfoliatin type A	Epidermolytic	Bullous impetigo, SSSS
	Exfoliatin type B	Epidermolytic	Bullous impetigo, SSSS

BACTERIA	TOXIN	TOXIN TYPE	CLINICAL DISEASE
	TSS toxin 1 (TSST-1)	Superantigen	TSS (menstrual > non-menstrual), food poisoning
	Staphylococcal enterotoxins A-C	Superantigen	TSS (non-menstrual > menstrual), food poisoning
<i>Streptococcus pyogenes</i>	Streptococcal pyrogenic exotoxins A, C	Superantigen	TSS (non-menstrual), scarlet fever

Abbreviations: SSSS = staphylococcal scalded-skin syndrome; TSS = toxic shock syndrome

TABLE 178-2 Gram-Positive Coccal Infections Associated with Toxin Production

Caused by *Staphylococcus aureus* and/or group A streptococcus (GAS) toxin-producing bacteria

Characteristic syndromes (all with skin manifestations), categorized by predominant clinical presentation:

Predominantly Cutaneous

- **Local:** Bullous impetigo
- **Generalized:** Staphylococcal scalded skin syndrome (SSSS)

Intermediate (Cutaneous and Systemic)

- **Local:** Recurrent toxin-mediated erythema (e.g., recurrent perineal erythema)
- **Generalized:**
 - Recalcitrant erythematous desquamating disorder
 - Other abortive hybrids of predominantly systemic manifestations

Predominantly Systemic

- **Generalized:**
 - **Toxic shock syndrome** — *Staphylococcus* (predominantly) or *Streptococcus*
 - **Scarlet fever** — *Streptococcus* (predominantly) or *Staphylococcus*